

Polar Curves Gallery: Identify & Sketch

Student: Albert Lee

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Part A ??Identify the Curve Family

A-Worked Example (classify $r = 2 + 2 \sin \theta$) ??solution shown in worksheet.

A1. Classify each equation

Lima??on

Rose

(a) $r = 3 + 5 \cos \theta$: lima??on, $a = 3, b = 5 \Rightarrow a < b \Rightarrow$ inner loop. Orientation: $+\cos \Rightarrow$ main lobe right.

(b) $r = 4 - 2 \sin \theta$: lima??on, $a = 4, b = 2 \Rightarrow a > b$ but $a = 2b \Rightarrow$ borderline convex (just at the convex threshold). Orientation: $-\sin \Rightarrow$ main lobe down.

(c) $r = 2 \cos 3\theta$: rose. $n = 3$ odd $\Rightarrow 3$ petals. First petal along $+x$ -axis.

(d) $r = 5 \sin 4\theta$: rose. $n = 4$ even $\Rightarrow 8$ petals. First petal along $\theta = \pi/8$ (22.5??).

a) inner-loop lima??on ?? b) convex lima??on (borderline) ?? c) 3-petal rose ??
d) 8-petal rose

A2. Lima??on orientation rule

Orientation Rule

(a) $r = 1 + \cos \theta$: right (max at $\theta = 0$).

(b) $r = 1 - \cos \theta$: left (max at $\theta = \pi$).

(c) $r = 1 + \sin \theta$: up (max at $\theta = \pi/2$).

(d) $r = 1 - \sin \theta$: down (max at $\theta = 3\pi/2$).

Rule (one sentence): The main lobe points in the direction where the trig term is most positive: $+\cos \rightarrow$ right, $-\cos \rightarrow$ left, $+\sin \rightarrow$ up, $-\sin \rightarrow$ down.

$\cos =$ horizontal axis (sign chooses L/R); $\sin =$ vertical axis (sign chooses U/D).

A3. Skeleton of $r = 2 \cos 5\theta$

Rose Anatomy

$n = 5$ odd $\Rightarrow$ **5 petals**, each of length 2. **cos** form means the first petal points along the positive x -axis. Petals are evenly spaced at $\theta = 0, 2\pi/5, 4\pi/5, 6\pi/5, 8\pi/5$.

5 petals, first along $+x$ -axis.

Part B ??Circles & Cardioids

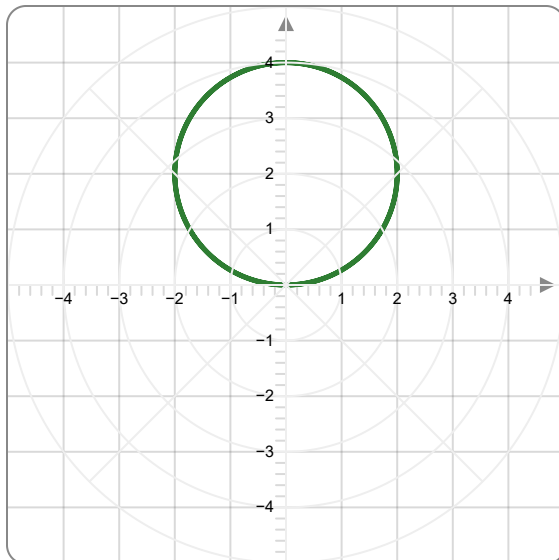
B-Worked Example ($r = 1 + \cos \theta$ cardioid right) ??solution shown in worksheet.

B1. $r = 4 \sin \theta$

Circle (sin)

Diameter 4

Circle of diameter 4 sitting on the positive y -axis. Center $(0, 2)$, radius 2. Passes through origin and $(0, 4)$.



Diameter 4, on positive y -axis.

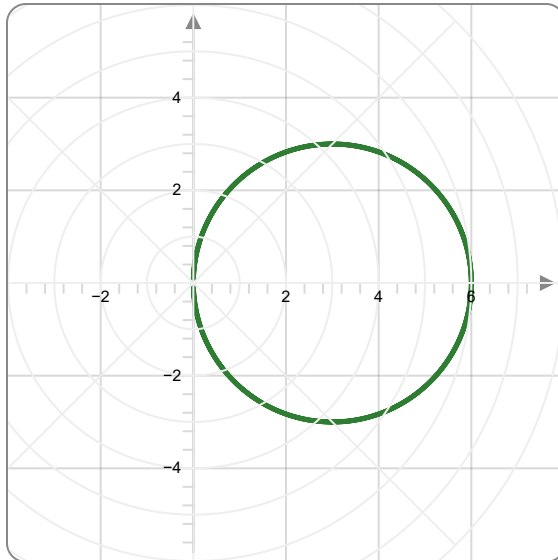
Circle, center $(0, 2)$, radius 2.

B2. $r = 6 \cos \theta$

Circle (cos)

Diameter 6

Circle of diameter 6 sitting on the positive x -axis (right of y -axis). Center **(3, 0)**, radius 3.



Diameter 6, right side of y -axis.

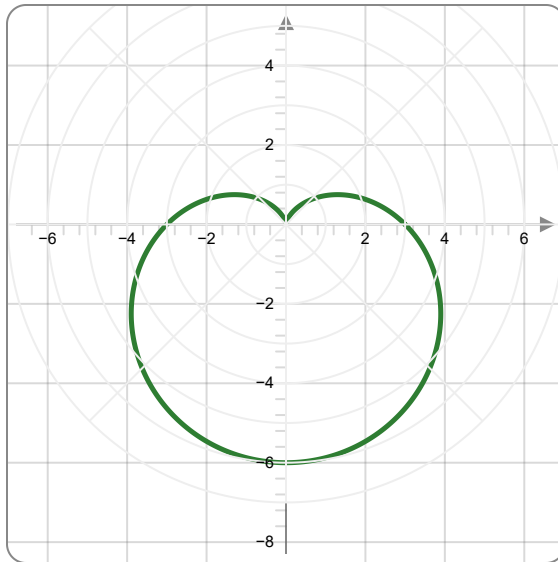
Circle, center (3, 0), radius 3.

B3. $r = 3(1 - \sin \theta)$

Cardioid

Downward

Cardioid. Max $r = 6$ at $\theta = 3\pi/2$ (bottom), $r = 0$ at $\theta = \pi/2$. Heart points **downward**.



Cardioid pointing down.

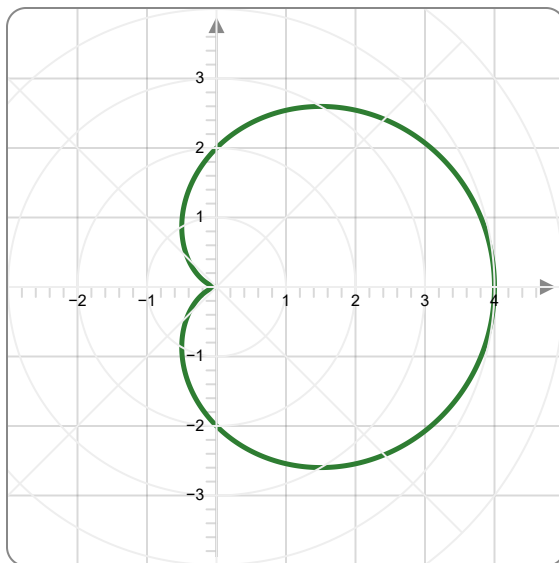
Cardioid, downward, max distance 6.

B4. $r = 2 + 2 \cos \theta$

Cardioid

Rightward

Cardioid. Max $r = 4$ at $\theta = 0$ (right), $r = 0$ at $\theta = \pi$. Heart points **right**. Mirror image of B3 (rotated 90° — sin vs cos swaps vertical ↔ horizontal).



Cardioid pointing right.

Cardioid, rightward, max distance 4.

Part C ??Limaçons

C-Worked Example ($r = 1 + 2 \sin \theta$ inner loop) ??solution shown in worksheet.

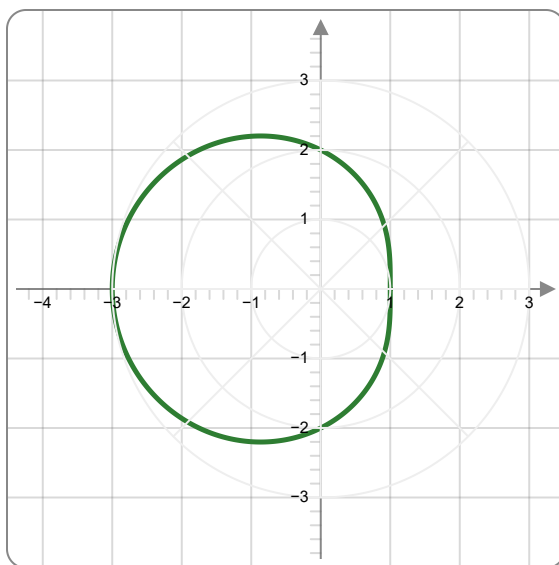
C1. $r = 2 - \cos \theta$

Dimpled Limaçon

Leftward

$a = 2$, $b = 1 \Rightarrow a > b$ but $a < 2b$, so **dimpled limaçon**. Sign $-\cos \Rightarrow$ max at $\theta = \pi$ (left), min (dimple) at $\theta = 0$ (right).

Key values: $\theta = 0 \Rightarrow r = 1$; $\theta = \pi/2 \Rightarrow r = 2$; $\theta = \pi \Rightarrow r = 3$; $\theta = 3\pi/2 \Rightarrow r = 2$.



Dimpled limaçon, oriented left.

Dimpled, left-pointing main lobe, max 3, dimple distance 1.

C2. $r = 1 - 2 \cos \theta$

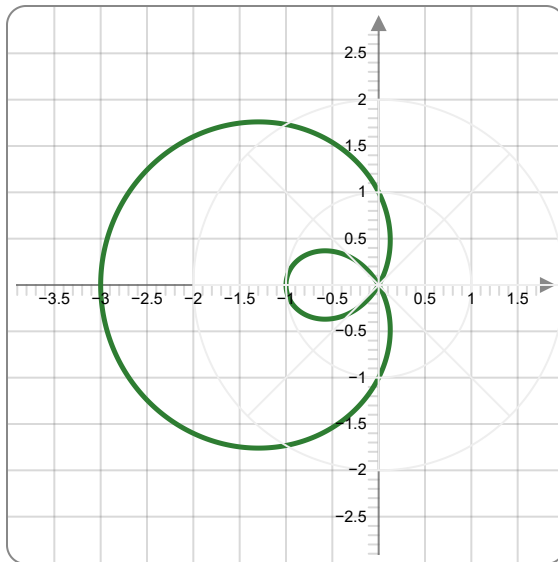
Inner-Loop Limaçon

Leftward

$a = 1$, $b = 2 \Rightarrow a < b$, so **inner-loop limaçon**. Sign $-\cos \Rightarrow$ main lobe points left.

Pole crossings: $1 - 2 \cos \theta = 0 \Rightarrow \cos \theta = 1/2 \Rightarrow \theta = \pi/3, 5\pi/3$. Inner loop between these angles, on the right.

Key: $\theta = 0 \Rightarrow r = -1$ (flips to left, distance 1 ??inner loop tip); $\theta = \pi \Rightarrow r = 3$ (main lobe tip, left).



Inner-loop limaçon, oriented left.

Inner loop with pole crossings at $\pi/3, 5\pi/3$; main lobe left.

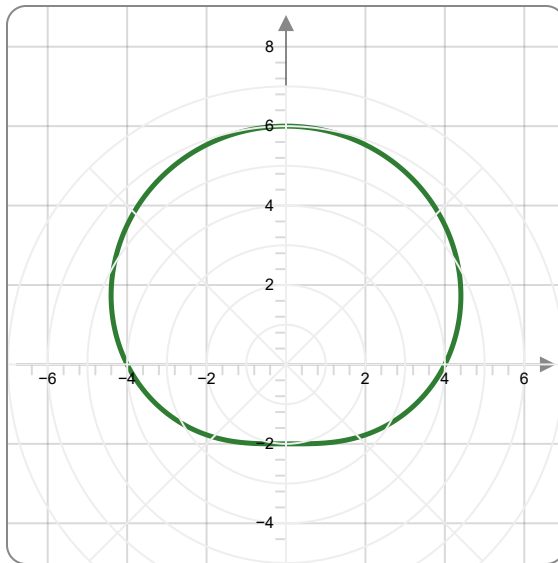
C3. $r = 4 + 2 \sin \theta$

Convex Limaçon

Upward

$a = 4$, $b = 2 \Rightarrow a = 2b$, exactly at the convex threshold. **Convex limaçon** (smooth oval, no dimple).

Max at $\theta = \pi/2$ ($r = 6$, top); min at $\theta = 3\pi/2$ ($r = 2$, bottom). Slightly elongated vertically.



Convex limaçon, elongated upward.

Convex oval, max 6 (top), min 2 (bottom).

Part D ??Roses

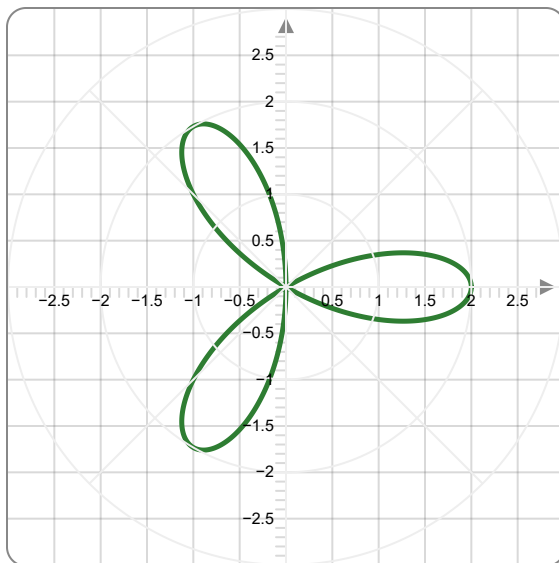
D-Worked Example ($r = 3 \sin 2\theta$, 4 petals) ??solution shown in worksheet.

D1. $r = 2 \cos 3\theta$

Rose

3 Petals

$n = 3$ odd $\Rightarrow$ 3 petals, length 2. **cos** form $\Rightarrow$ first petal along $+x$ -axis ($\theta = 0$). Other two at $\theta = 2\pi/3$ and $\theta = 4\pi/3$.



3-petal rose, first along $+x$.

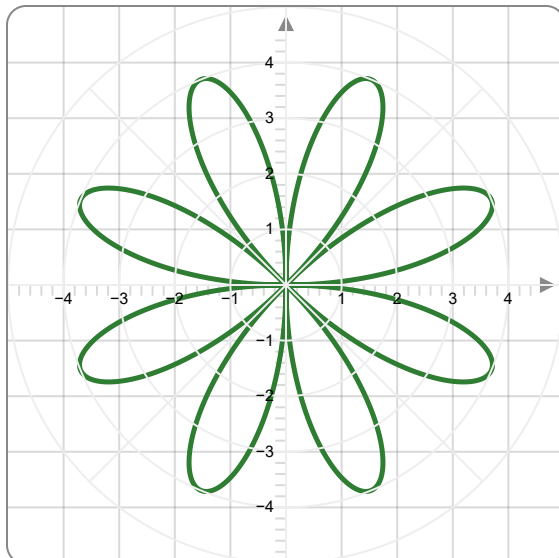
3 petals, length 2, evenly spaced 120° apart.

D2. $r = 4 \sin 4\theta$

Rose

8 Petals

$n = 4$ even $\Rightarrow 2n = 8$ petals, length 4. **sin** form $\Rightarrow$ first petal along bisector
 $\theta = \pi/(2 \cdot 4) = \pi/8$ (22.5째). Petals every $\pi/4 = 45$ 째 thereafter.



8-petal rose, length 4.

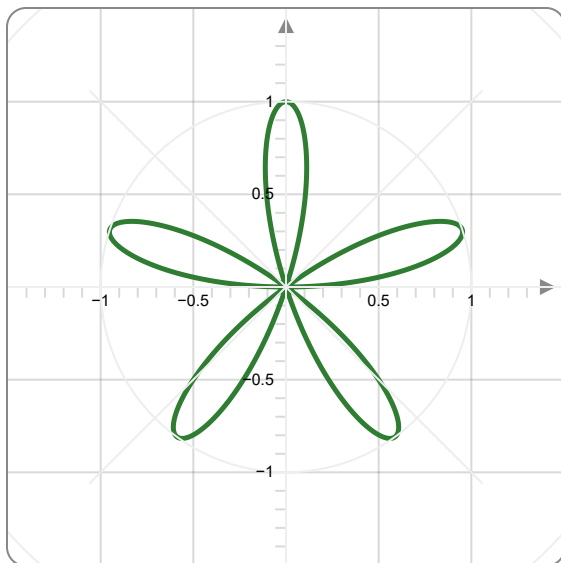
8 petals, length 4, spaced 45째 apart, first at 22.5째.

D3. $r = \sin 5\theta$

Rose

5 Petals

$n = 5$ odd $\Rightarrow$ 5 petals, length 1. **sin** form $\Rightarrow$ first petal along $\theta = \pi/10$ (18째). Petals at $\pi/10, \pi/10 + 2\pi/5, \dots$ (every 72째).



5-petal rose, length 1.

5 petals, length 1, spaced 72째 apart, first at 18째.

Part E ??Edge Cases

E-Worked Example ($r^2 = \tan \theta$) ??solution shown in worksheet.

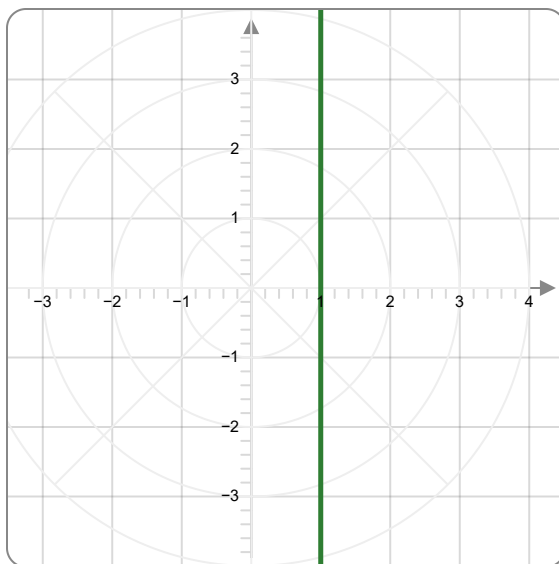
E1. $r = \sec \theta$

Vertical Line

Polar?Subject

Rewrite: $r = \frac{1}{\cos \theta} \Rightarrow r \cos \theta = 1 \Rightarrow x = 1$.

This is the **vertical line $x = 1$** .



Vertical line $x = 1$.

$x = 1$ (vertical line).

E2. $r = \frac{4}{1 + 2 \sin \theta}$

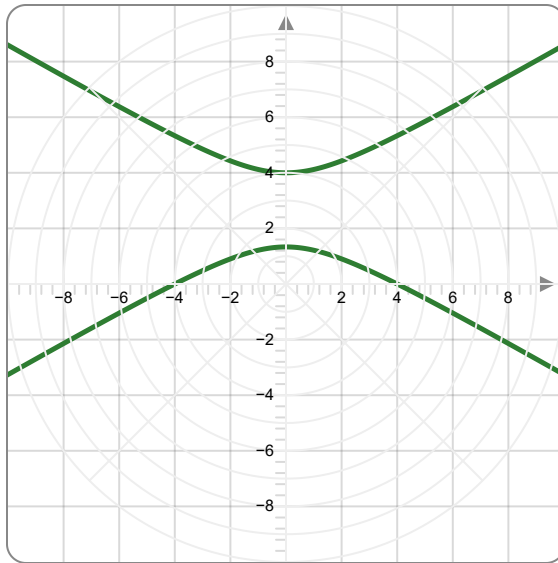
Conic (Hyperbola)

Eccentricity 2

Conic section in polar form with eccentricity $e = 2$. Since $e > 1$, this is a **hyperbola**.

Denominator zero when $\sin \theta = -1/2 \Rightarrow \theta = 7\pi/6, 11\pi/6$. At these angles $r \rightarrow \pm\infty$ the asymptotic directions of the hyperbola.

At $\theta = \pi/2$ (top): $r = 4/3$. At $\theta = 3\pi/2$ (bottom): $r = 4/(-1) = -4$ (flips to top, distance 4). At $\theta = 0$: $r = 4$. At $\theta = \pi$: $r = 4$.



Hyperbola, eccentricity 2.

Hyperbola ($e = 2$) with asymptotic angles at $\theta = 7\pi/6, 11\pi/6$.